

JINDAL SAW LIMITED • SEAMLESS PIPE DIVISION

MILL TEST CERTIFICATE — EN 10204 3.1 INSPECTION CERTIFICATE

Customer:	Indian Oil Corporation Ltd	Cert No:	MTC-JSL-2021-44719
Product:	Seamless Carbon Steel Pipe	Date:	12-MAR-2021
Specification:	ASTM A106 / ASME SA106 Grade B	Heat No:	4471
Dimensions:	12" NPS (323.8 mm OD) x Sch 40 (9.53 mm WT)	Lot / Bundle:	B-8841-A

1. CHEMICAL COMPOSITION (HEAT ANALYSIS %)

Element	C	Mn	P	S	Si	Cr	Cu	Ni	Mo	V
Actual %	0.21	0.88	0.012	0.008	0.22	0.08	0.12	0.06	0.02	0.01
Code Max	0.30	1.06	0.035	0.035	0.10 min	0.40	0.40	0.40	0.15	0.08

2. MECHANICAL PROPERTIES & TESTS

Property Tested	Specified Requirement (ASTM A106 Gr B)	Measured Value	Result
Tensile Strength	min 415 MPa (60,000 psi)	485 MPa (70,340 psi)	PASSED
Yield Strength	min 240 MPa (35,000 psi)	310 MPa (44,960 psi)	PASSED
Elongation in 2 in	min 30.0 %	34.5 %	PASSED
Hydrostatic Test	min 2,500 psig (hold 5s)	2,850 psig (10s hold)	PASSED
Flattening Test	Per ASTM A530 Section 10	No cracks observed	PASSED

CERTIFICATION: We hereby certify that the material described above has been manufactured, sampled, tested, and inspected in accordance with specification ASTM A106 / ASME SA106 Grade B and satisfies all mandatory physical, chemical, and hydrostatic requirements.

Quality Assurance Manager:	Dr. K. N. Rao	Authorized Inspection Stamp:
Signature / Digital Seal:	K.N. Rao (Digitally Signed)	[GOVERNMENT APPROVED NDT AGENCY]